
Statecraft Before Capability: Aspirational Security and State Competition over Not-Yet Technologies

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Abstract

How do states conduct statecraft over capabilities that do not yet exist? Scholarship on economic statecraft and technological competition generally assumes a material object, such as a chokepoint, a supply chain, or an already deployed system that can be controlled, denied or leveraged. Yet states increasingly commit public investment, build institutions and stake sovereignty claims in domains that remain pre-material, uncertain and largely undeployable. This article introduces the concept of *aspirational security*: a form of anticipatory state behaviour in which governments compete for strategic position in not-yet-realised technological domains. It proposes a triadic model composed of anticipatory governance, strategic narratives and status-seeking behaviour, which together seeks to explain how states construct technological futures, narrate sovereign identity, and claim relevance in an order shaped by competition over technologies that do not yet work, or not even exist.

Departing from analyses centred on great-power rivalry, the study examines seven globally influential states below the great-power threshold — the United Kingdom, France, Germany, Canada, Japan, South Korea and India — arguing that these actors, precisely because of their material constraints, rely more heavily on symbolic projection and future-claiming. Their national quantum strategies offer an ideal empirical domain to explore anticipatory statecraft of this kind: unlike AI, quantum technologies remain largely pre-deployment, yet are routinely framed as immensely geopolitically transformative. The empirical analysis reveals convergence in anticipatory governance but striking variation in narrative and prestige dimensions, with the vocabulary of technological sovereignty and strategic autonomy functioning as anticipatory positioning rather than as a factual description of capability. The article introduces a typology of strategic profiles and advances a dynamic theory of aspirational strategy formation under technological immaturity, offering a generalisable model for analysing statecraft in an era when the future itself has become a site of contestation.

1 Introduction

How do states compete over technologies that are not yet fully real? How do aspirational technologies become sites of geopolitical contestation before they even “exist”? In recent years, governments around the world have launched ambitious strategies for emerging technologies whose capabilities remain speculative and whose timelines are uncertain. From artificial general intelligence (AGI) and nanotechnology to quantum computing, states are increasingly engaging in policy coordination, public investment, international signalling, and discursive framing long before these technologies reach the levels of maturity or deployability that would allow for their integration with existing capabilities. These strategic behaviours pose an important puzzle: how do states compete over technologies that are not yet materially real?

Nowhere is this dynamic more visible—and more puzzling—than in the case of quantum technologies. Encompassing advances in computing, communication, and sensing based on the exploitation of the quantum-mechanical phenomena of superposition, entanglement and tunnelling, these technologies are widely portrayed as so transformative that they will amount to paradigm-shifting game-changers for national security, economic competitiveness, and technological sovereignty. Yet despite their prominence in strategy documents and the significantly increased media attention they have attracted, most quantum technologies remain in early research phases. Quantum computers are far from breaking modern encryption, quantum communication systems are still largely experimental, and quantum sensors have not yet reshaped military intelligence or navigation. Nevertheless, for some time, states have been articulating grand ambitions, have committed substantial funding, and positioned themselves symbolically in an accelerating “global quantum race”.

I argue that national quantum technology strategies are not merely industrial roadmaps but are best understood as aspirational security performances. Specifically, they function as narratives to lock in not-yet technologies—publicly staged scripts through which states seek to conjure future sovereignty, project leadership in an uncertain international system, and assert epistemic authority over emerging technological domains. By reading these strategies as anticipatory security artefacts as well as symbolic interventions, this article seeks to better understand how states manage techno-ontological insecurity amid rapid technological change.

National quantum strategies reflect a broader transformation in the strategic logic of state competition in the emerging technology space. In contrast to traditional models of competition over material capabilities, states now engage in what I call *aspirational security behaviour*: a mode of geopolitical action in which governments compete over symbolically charged technological futures, rather than immediately deployable capabilities. Aspirational security behaviour is anticipatory, narrative-driven, and status-sensitive. It is oriented toward projecting sovereign futures, claiming leadership over anticipated technological capabilities, and shaping strategic identities in an increasingly multipolar world.

To explain this behaviour, the article introduces a triadic model of strategic competition over emerging technologies. The model identifies three interlocking components:

1. **Anticipatory Governance:** States act in the present to shape and prepare for expected technological futures, establishing long-term institutions, funding frameworks, and regulatory architectures in the absence of operational capabilities.
2. **Strategic Narratives:** States construct coherent discursive framings to render those futures politically intelligible and legitimate. Through the language of threat, sovereignty, independence, and leadership, these narratives link emerging technologies to national identity and global role claims.
3. **Status-Seeking Behaviour:** States symbolically position themselves within global hierarchies of technological prestige. By seeking or claiming leadership in emerging fields, they seek recognition and relevance in an international system increasingly shaped by perceptions and expectations of innovation power.

Together, these three components constitute a distinct mode of international competition that is both material and performative, forward-looking, and shaped by both institutional design and symbolic projection grounded in actual engineering capabilities and capacity-building: to be successful in the exercise of aspirational security posturing, states will require a minimum of actual technological capacity, infrastructure and skilled labour to make sufficient progress in relevant domains. Moving beyond realist models of material balancing or liberal theories of institutional coordination, this model accounts for the ways in which emerging technologies become arenas of geopolitical positioning well before their technical realisation.

The article grounds this theory in a comparative analysis of national quantum strategies from seven technologically advanced and aspirational states that seek to keep up with the United States and China in international technology competition: the quantum strategies of the United Kingdom, France, Germany, Canada, Japan, India, and South Korea provide the empirical base of this study. These states were selected for their high research and development capacity, their active engagement in quantum policymaking, and their constrained geopolitical status relative to the US and China.

As “states of global influence”, avoiding the contested term “middle powers” ([Elliott 2024](#)), they seek their place in a multipolar world. Yet their military and technological capabilities are subpar relative to the US and China so that they are particularly reliant on symbolic action, discursive leadership, and future-oriented governance to stake claims in advanced technologies that promise to have a seismic impact on defence and security. While these countries differ in region, political structure, and institutional makeup, each of these states is attempting to secure their share of a new technology their quantum strategies describe as intensely transformative—one day.

Using a theory-driven, index-based coding frame derived from the triadic model, the article traces how each state engages in anticipatory governance, crafts strategic narratives, and signals status through its quantum ambitions. The empirical analysis reveals notable variation in how states perform aspirational security. While anticipatory governance registers consistently high scores across all seven countries, significant divergence emerges in the narrative and status-seeking dimensions. The UK and France adopt high-intensity strategies that combine strong governance with bold identity framing and international prestige claims. Germany, by contrast, focuses heavily on institutional infrastructure while downplaying symbolic projection. India and South Korea exhibit more uneven profiles, with strong rhetorical emphasis but weaker implementation or coordination mechanisms. Japan presents a hybrid case, pairing extensive governance and status-seeking efforts with a more muted narrative framing. These patterns suggest that while states converge on the need to prepare for quantum futures, they differ systematically in how they narrate, govern, and signal their strategic quantum ambitions. Such variation reflects deeper institutional, cultural, and geopolitical logics, which I categorise through a typology of aspirational strategy profiles.

Empirically, the article contributes to the comparative study of states below the great-power threshold in emerging technology governance. Much of the literature on quantum geopolitics focuses on the US and China, treating other actors as marginal.

Yet, as this analysis shows, states of global influence below the US/China benchmark are active participants in shaping the symbolic, institutional, and strategic contours of quantum technology futures. In doing so, they demand an expansion of scholarly inquiry beyond binary superpower rivalry, and thus invite research into diversified forms of technology leadership and sovereignty projection. Theoretically, the triadic model developed in this paper offers a novel mid-range theory for understanding *how states engage in pre-material competition*, a phenomenon that is becoming increasingly central to global politics.

The remainder of the article is structured as follows. Section 2 reviews existing literature on emerging technologies, international competition, and strategic behaviour, identifying the gap that aspirational security addresses. Section 3 develops the triadic theoretical framework, defining its three components and explaining its relevance under conditions of technological immaturity. Section 4 presents the research design, introducing the scoring methodology, coding procedure, and formalisation of the Aspirational Security Intensity (ASI) index. Section 5 reports the empirical findings in four stages: a cross-dimensional comparison, dimension-by-dimension analysis, typological clustering, and visualisation of strategic variation. Section 6 concludes by reflecting on the implications of the model, its scope conditions, and avenues for future research in the geopolitics of emerging technologies. A detailed Appendix follows, containing country-level scoring matrices and explanatory notes for all seven strategies.

2 Conceptual and Theoretical Positioning

The strategic competition over emerging technologies has become a defining feature of international relations in the twenty-first century (Allenby 2014, Cheung 2022, Lin 2023, Sechser et al. 2019). From AI to advanced semiconductors, and more recently, quantum technologies, global powers are increasingly engaged in efforts to secure dominance in future-defining domains, prompting a scholarly shift of attention and expansion beyond traditional weapons systems and alliances, toward the politics of innovation, infrastructure, and digital control (Csernatoní & Martins 2024, Sechser et al. 2021, Haddad et al. 2024). These newly emerging technologies challenge traditional security frameworks by blurring the lines between civil and military applications (Vaynman & Volpe 2023), lowering barriers to entry for non-state actors, and accelerating the pace of strategic change (Broeders & van den Berg 2020,

[Lindsay 2020](#)).

States “weaponise” economic interdependencies, oftentimes arising from deep technological integration, not just for economic gain but coercive diplomacy and strategic purposes ([Farrell & Newman 2019](#)). This shift has made technologies instruments of geopolitical competition rather than merely economic assets. Advanced technologies enable leveraging countries to gain “undue influence” and exploit technological dependencies for strategic gain and advantage ([Drezner et al. 2021](#)). Yet, much of the literature in this domain continues to focus on capabilities that are already real and long-established, or near deployment, or adopt a hypothetical position *as if* the technology in question were fully real already. This move seems an accepted and largely unquestioned workaround given that any new technology “with less empirical evidence than is available for technologies that have been around for decades, also generates methodological challenges for research” ([Horowitz 2020](#)).

Concurrently, policymakers and officials are increasingly committed to future-facing technology forecasting and anticipatory policy and rules-making. Across numerous countries, governments have established dedicated agencies, task forces, or initiatives focused on emerging technologies and their anticipatory regulation. Examples include the establishment of, for instance, AI Safety Institutes in the US and the UK. These bodies are often tasked with monitoring technological developments, assessing their potential implications, and developing targeted governance models. Research on AI policy, for instance, suggests that policymakers pre-emptively respond to public concerns about plausible future AI harms that have not yet materialised ([Schiff 2024](#)).

Increased emphasis on foresight and horizon-scanning activities across governments involves the structured analysis of potential future technological developments, their timelines, and their likely societal, economic, and security implications. Organisations such as the UK’s Government Office for Science, or the Digital Regulation Cooperation Forum and various think tanks play a crucial role in these efforts. This has led to the development of early-stage regulatory frameworks, ethical guidelines, and discussions around responsible innovation in areas like AI, gene editing, and autonomous systems ([Kalenzi 2022](#), [Kreps 2024](#)). As a result, a growing disjuncture has emerged between the temporal horizons of strategists and policymakers in government who become increasingly skilled at “futuring” emerging technologies, as it is expressed in efforts at institutionalising the horizon-scanning of not-yet technologies, and the conceptual tools available to researchers to situate and explain such temporal, techno-strategic positioning.

This article responds to this disjuncture by offering a model of aspirational security behaviour, one that accounts for how states engage in geopolitical competition over technologies that remain speculative, incomplete, or are “not quite there yet”. While previous work has explored aspects of this phenomenon—through the lenses of anticipatory governance and regulation, status-seeking, or discursive construction—this paper integrates these conceptual frames to explain how these elements interact. This section reviews the key bodies of literature that inform the triadic model developed in this article, highlighting their contributions and limitations in accounting for emerging technology competition in a pre-material state, defined as a space of great uncertainty where engineering progress on the ground has not yet translated into capabilities that would make a material difference to the decision portfolios of state actors.

2.1 Strategic Competition and Emerging Technologies

From the offence-defense balance to tipping points in military affairs, scholars and practitioners have displayed growing interest in the mechanisms by which new technologies may alter state behaviour, impact threat perception, and erode strategic stability ([Altmann & and Sauer 2017](#), [Berghofer et al. 2023](#), [Fetter & and Sankaran 2025](#), [Garfinkel & and Dafoe 2019](#), [Montgomery & and Nelson 2025](#), [Sechser et al. 2019](#), [Warren 2025](#)). Technological progress by one state may trigger responses from strategic competitors that ultimately reduce security for all parties, creating escalatory dynamics similar to traditional security dilemmas ([Steff et al. 2021](#)). Technological competition is motivated by the pursuit of relative advantage in an anarchic international system in which institutions and the liberal order that backs them are at risk of disintegrating ([Mearsheimer 2019](#)).

Against such currents, recent work has zoomed in on digital, autonomous, and computational technologies, including AI, cyber capabilities, and now quantum systems, with a view to strategic US-China competition ([Bresnick 2024](#), [Broeders et al. 2019](#), [Fedasiuk & Weinstein 2022](#), [Kerbusch et al. 2018](#), [Konaev et al. 2023](#), [Krause 2024](#), [Saylor 2020](#)). The real or perceived impact of activities in cyberspace has gained particular attention, from early-day excitement about the allegedly seismic fallout of cyber operations ([Dunn Cavelty et al. 2024](#)) to viewing them as yet another “structural modifier” by which states compete within a structurally given system ([Foulon & Meibauer 2024](#)). Much of this literature emphasises the (perceived or expected)

instrumental value of technologies in achieving military superiority, shaping deterrence postures, or enabling new forms of coercion. Matching many policy officials' curiosity, much scholarly attention is skewed towards the potentially offensive qualities and properties of emerging technologies, leaving their plausible roles in defensive posturing somewhat underexplored ([Loneragan 2024](#)). Emerging technologies that favour offence over defence may increase security competition and instability ([Gartzke & Lindsay 2015](#)).

In the case of quantum technologies, researchers have highlighted their potential to disrupt cryptographic systems, enhance sensing and targeting capabilities, and increase the operational tempo of military logistics and decision-making ([Choi 2023](#), [Krelina 2021](#), [Lele 2021](#)). These accounts make a compelling case for why quantum matters—eventually. Yet, they often treat state behaviour as a rational response to tangible capabilities, or to clear signals of technological maturity, or outright operate from an imagined future when these capabilities are fully realised. Conceptual moves of this sort invite inquiring much more closely how and why states invest symbolically and materially in technologies that remain experimentally constrained, costly, and operationally distant.

It is important to note that strategic competition in the above discourse is often framed in binary terms: the US and China are cast as the dominant actors, while other states are relegated to the role of late movers, followers, or de facto rule-takers. This risks neglecting the ways in which powers with (bounded) global influence seek to shape emerging technology orders, not through overwhelming material capabilities, but through symbolic positioning, anticipatory planning, and narrative construction. These behaviours require analytical tools that go beyond traditional models of capability-based rivalry.

2.2 Technological Sovereignty

One approach to the governance of not-yet technologies that has gained significant traction in both policy and academic circles is the concept of “technological sovereignty”. Defined loosely as the ability of a state or region to develop, maintain, and control critical technologies, technological sovereignty has become a rallying cry for the European Union and several of its member states. In 2019, newly elected European Commission President Ursula von der Leyen outlined it as achieving “mastery and ownership of key technologies in Europe”, including quantum

computing, artificial intelligence, blockchain, and critical chip technologies (Commission 2019). The European Parliament identified quantum as one of several “key enabling technologies” essential for Europe’s future, to be tightly aligned with its industrial strategy. It defined (quantum-)technological sovereignty as “the ability for Europe to develop, provide, protect, and retain critical technologies required for the welfare of European citizens and businesses and to act and decide independently in a globalised environment” (European Parliament et al. 2021). Parallel to the diffusion of this term, the concept of “digital sovereignty” has become a central tenet of the EU’s cyber agenda. Ultimately, this is to promote “strategic autonomy”, yet another notion that gained traction since its introduction in the 2016 European Global Strategy. This strategy seeks to bolster the EU’s ability to act (more) independently, given American and Chinese dominance in advanced technologies.

The terms “digital sovereignty”, “technological sovereignty”, and “strategic autonomy” are often used interchangeably, hampering the EU’s ability to formulate and communicate cohesive strategies. The scope of European strategic autonomy has expanded to various policy areas, often fusing the term with similar concepts like “open strategic autonomy” and “strategic sovereignty”, which seems to have diluted policy focus (Damen 2022). The widely attested “strategy cacophony” (Meijer & Brooks 2021) in Europe seems to be complemented by “concept cacophony”. While Europe may be lacking in global Big Tech corporations, it seems world-leading in the deployment of aspirational, discursive strategies that invite closer conceptual attention.

While the EU has been successful in crafting regulatory frameworks to protect its internal market from third-country digital intermediaries, commentators find that this approach does little to establish true sovereignty in the field of emerging technologies like AI—European efforts are seriously constrained by a lack of a robust technological base, limited defence strategy, and an over-reliance on regulatory measures (Calderaro & Blumfelde 2022), pointing to the immense status of anticipatory regulation in European policymaking. While regulatory measures, such as the General Data Protection Regulation (GDPR) and the Digital Services Act (DSA) shape global standards, they do not necessarily foster technological innovation within the EU. For advanced security and defence technologies, Europe commits comparably less R&D investment relative to international competition (Csernatoni 2022, 2021). With limited capacity to develop and deploy its own technological solutions, curiously Europe has become a strong rule-maker without the proper means to compete effectively in the very arena it seeks to govern (Floridi 2020).

What remains underexplored in the literature that critically reflects on the distance between discursive strategy and industrial readiness, however, is the symbolic and anticipatory function of such sovereignty claims. As the empirical analysis in later sections demonstrates, sovereignty in quantum strategies especially is as much a narrative performance as it is a practical objective. States use the language of sovereignty to signal resolve, attract investment, and define their position relative to allies and competitors. The decidedly performative dimension of technological sovereignty is difficult to capture in existing models focused on capacity or regulation. It requires a theory that treats technological sovereignty as an aspirational narrative, rather than a binary condition, yet at the same time avoids the pitfalls of constructivist reduction to mere “speech act” or “ideology” with no material grounding.

2.3 Anticipatory Governance: Commanding the Not-Yet

Anticipatory governance refers to a systematic approach that enables organisations and governments to prepare for and manage emerging challenges proactively. It aims to integrate foresight and horizon-scanning to navigate complexities before they manifest fully. Anticipatory governance, therefore, refers to the capacity of institutions to engage with uncertain futures, particularly in areas where technical developments are not yet fully understood or realised ([Jasanoff 2016](#), [Jasanoff & Kim 2015](#)).

The concept alludes to the risks of technologies developing too fast, or diffusing too rapidly, for stakeholders to keep up, e.g. in its definition as a “broad-based capacity [...] to manage emerging knowledge-based technologies while such management is still possible” ([Guston 2014](#)). Efforts typically include the use of foresight, public engagement, scenario planning, and early policy interventions to shape the trajectory of emerging technologies ([Barben et al. 2007](#), [Fuerth 2009](#)). Intellectually, it demands an openness and curiosity on the part of policymakers and practitioners about the not-yet and high levels of “future receptivity” ([Heo & Seo 2021](#)).

In 2024, the OECD published a “Framework for Anticipatory Governance of Emerging Technologies”, pointing to the dual character of most of them as at once promising unprecedented progress as well as carrying substantial risks. It recommends a series of tools “to anticipate and get ahead of governance challenges and build longer-term capacities to shape innovation more effectively” ([OECD 2024](#)). Climate change policies have attracted research into anticipatory governance models ([Boyd et al. 2015](#)). Finland’s exploration of anticipatory innovation governance is an example

of governmental efforts to foster more adaptive and forward-thinking policy environments ([Toenurist 2022](#)).

Although primarily applied to domestic policy contexts—such as nanotechnology or synthetic biology, in addition to the cases above—as a concept, anticipatory governance invites connections with international security. While some work is underway with a view to applying the concept to international efforts at harmonising the regulation of AI ([Taeihagh 2025](#), [Umbach 2024](#)), anticipatory logics in international affairs beyond 2x2 payoff calculations remain an under-researched topic.

This article extends anticipatory governance into the geopolitical domain, *where uncertainty is not merely a risk to be managed, but increasingly a terrain of competition in itself*. In the case of quantum technologies, national strategies often display such anticipatory characteristics: they institutionalise long-term funding, build consortia for building testbeds, and create central agencies to steward future quantum capabilities. These practices are not reactive; they are future-claiming acts. When read through a geopolitical lens, anticipatory governance becomes a mechanism through which states project technological sovereignty forward in time—and in competition with others.

2.4 Strategic Narratives

A second component of aspirational security behaviour, I argue, is strategic narration. “Strategic narratives” are frameworks constructed by political actors to articulate and promote their agendas and policies within the international arena. They serve as tools to shape perceptions, influence behaviours, and manage expectations among domestic and international audiences. In a popular definition, strategic narratives are “tools that political actors employ to promote their interests, values, and aspirations for the international order by managing expectations and altering the discursive environment” ([Miskimmon et al. 2017](#)). They define and communicate “who we are” and “what kind of world order we want” (*ibid.*).

The concept of strategic narratives intersects with theories of “soft power” ([Nye 2005, 2019](#)) and international communication. It emphasises the role of storytelling in constructing social realities and influencing international relations, “stressing the role played by storylines in affecting public attitude” ([Coticchia & Catanzaro 2022](#)). Narratives of this type have been analysed across various contexts in international relations, especially regarding great-power identity: narratives are instrumental in

shaping and projecting the identities of leading powers, influencing how they are perceived globally. While strategic narratives are powerful tools, their relationship with soft power is complex, requiring significant boundary work to differentiate the term from other high-level abstractions such as “frame”, “master narrative” and “ideology” (Coticchia & Catanzaro 2022). Ultimately, the concept seeks to uncover how narratives influence domestic and international audiences, and how these audiences interpret state behaviour, especially in ambiguous or contested contexts.

Emerging technologies are fertile ground for narrative construction. Because their futures are uncertain, their meanings are politically malleable. States can frame quantum computing, AGI, or nanotechnology as engines of freedom, threats to values, instruments of leadership, or symbols of decline. Strategic narratives do not merely accompany not-yet technologies but co-constitute them. They provide the rhetorical logic through which investment, coordination, and competition become justifiable.

As will be explored in this article, national quantum strategy documents reveal distinct narrative styles: for instance, the UK foregrounds its status as a “science and technology superpower” in a post-Brexit context, France emphasises strategic autonomy and future conflict readiness, while Germany frames its efforts as contributions to collective European sovereignty. These narratives reflect deeper political identities and histories, but they also do important heavy lifting in the present: they mobilise domestic support and secure legitimacy, attract external recognition, and structure international alignment.

2.5 Status-Seeking and Symbolic Competition

Status-seeking behaviour represents a fundamental dimension of international relations, driving state actions beyond traditional material power calculations (Götz 2021, MacDonald & Parent 2021, Røren 2024). States are seldom purely rational utility maximisers, but also status-conscious actors deeply invested in international recognition, reputation, and relative position (Paul et al. 2014). Status-seeking, therefore, refers to the efforts of states to attain a favourable position within the global hierarchy, aiming for respect and prestige from other international actors (Dafoe et al. 2014, Renshon 2017). This pursuit is driven by the inherent desire of states to enhance their standing and influence in the international system. Status considerations can be pivotal in shaping state behaviour and foreign policy decisions.

Smaller states, and countries of limited global influence may seek status to move up in the international hierarchy “by being recognised as a ‘good’ state” (Wohlforth et al. 2018). For instance, they may “perform” humanitarianism and goodness as part of larger signalling efforts (Bilgic 2024), or experience a “status boost” following the demise of an oppressive regime, e.g. South Africa’s much reversed international standing following the end of apartheid (Wohlforth 2025). States may pursue recognition, prestige, and respect as ends in themselves, independent of, or in addition to material gains (Beaumont & Røren 2025, Larson & Shevchenko 2010). While traditionally applied to the signalling of values, diplomatic behaviour, norm entrepreneurship, or international institutional leadership, status-seeking is increasingly visible in the realm of technological competition.

Emerging technologies offer states new opportunities to perform leadership to an international audience, even if they cannot dominate militarily or economically. “Prestige races” over emerging technologies motivate significant R&D spending (Barnhart 2022). Powers with bounded global impact in particular use technology policy to signal relevance, gain agenda-setting influence, and shape perceptions of being a force to be reckoned with. Claims to be the first to build a quantum computer, the best place for quantum startups, or the most favourable regulatory environment, are all status claims, not merely economic goals. The first nation to send astronauts to the moon, NASA’s rocket engineering feats in the 1960s emboldened the US, and helped solidify global perceptions of America as the leading technological power for decades to come.

I extend the logic of status-seeking to a pre-material domain. In doing so, I argue that symbolic projection is not secondary to strategy but is co-constitutive of it. Aspirational technologies are not just means to power in the future, they are stages for present status performance.

3 Theoretical Framework: A Triadic Model of Aspirational Security Behaviour

While each of the literatures discussed above explains aspects of the behaviour observed in the quantum technology case, none offers a complete account of how and why states engage in strategic competition over not-yet technologies. By integrating anticipatory governance, strategic narratives, and status-seeking into a single framework and dynamic model, this article aims for a more comprehensive and generalisable scheme for understanding emerging technologies vis-a-vis strategic competition. In what follows, I develop a triadic model of aspirational security behaviour, specifying its conceptual components, scope conditions, and empirical indicators. This model does not seek to replace existing theories of technological competition, but complements them by explaining how symbolic, narrative, and institutional dynamics interact in the governance of pre-material emerging technologies under strategic uncertainty. The framework responds to the shift from states reacting to the present to *claiming the future as a geopolitical space*.

3.1 Aspirational Security Behaviour

Aspirational security behaviour refers to state actions aimed at locking in desirable technology futures through pre-material competition. It is aspirational in two senses: first, because it pertains to technologies that are not yet fully realised; second, because it involves projecting sovereign ambition and technological identity into an uncertain future. The central claim of this article is that in wrestling with pre-deployable, experimental technologies, states act well before levels of technical maturity will offset uncertainties about the technology in question. Instead, they engage pre-emptively in strategic behaviours designed to shape perceptions, institutions, and narratives ahead of material capability.

This form of behaviour is especially relevant for powers with global influence that are not great powers—states. They can leverage substantial scientific and technological capacity, but lack the hard-power resources or global agenda-setting authority of the US or China. For such actors, aspirational security becomes a way of remaining visible and relevant in competitive technological fields. It allows them to position themselves not as passive adopters of innovation, but as active agents that (re)shape the global order.

3.2 Component 1: Anticipatory Governance

The first component of the model is anticipatory governance, defined above as the institutional capacity to prepare for, shape, and intervene in the management and regulation of technologies that are not yet operational. As discussed in the previous section, anticipatory governance involves foresight practices, long-term planning, institutional design, and pre-emptive regulatory effort. In international politics, it takes on a strategic dimension: states act in the present to pre-structure future technological trajectories, align resources with expected breakthroughs, and reduce their dependence on external actors.

Anticipatory governance is incorporated into the triadic model of aspirational security not as a prescriptive tool for policymakers, but as a diagnostic lens to identify and interpret patterns of state behaviour. National quantum strategies are replete with ambitious timelines, symbolic language, and institutional innovation, despite the fact that it has been more than four decades since the initial conception of a theoretical quantum computing device. To understand these documents as more than rhetorical artefacts or economic planning tools, a framework is required that *foregrounds the politics of the not-yet*.

Assessed in tandem with strategic narratives and status-seeking behaviour, quantum technology strategies reveal a coherent pattern: states are not merely reacting to global technological developments but are actively projecting themselves into an emerging strategic domain, staking claims in a future order that is still under construction. I argue that states recognise uncertainty as a productive space in which they can act, signal, and compete.

In the context of quantum technologies, anticipatory governance is visible in national strategies that lay out 10–15 year timelines, establish new coordinating bodies, allocate funding across R&D phases, and define future use cases. It also includes participation in multilateral roadmaps, development of post-quantum cryptographic standards, and efforts to establish domestic supply chains. These actions are often undertaken despite significant uncertainty about which of the many experimental quantum technologies will succeed, if any, how long they will take to materialise, or what their precise applications will be. As such, anticipatory governance reflects a willingness to invest in ambiguity, converting uncertainty into a domain of action.

3.3 Component 2: Strategic Narratives

The second component is strategic narratives: the representational and discursive frameworks through which states explain, justify, and make legible their technological ambitions.¹ As discussed above, strategic narratives are understood here as tools of identity construction and audience persuasion. In contexts of technological uncertainty, narratives serve to stabilise meaning and mobilise support for investment, institutional change, and international signalling.

Incorporating strategic narratives into a triadic model enables the identification of how states are utilising techno-storytelling to assert their roles and aspirations. Such narratives are multivalent in audience. Domestically, they offer reassurance, legitimacy, and a sense of direction amid uncertainty. Bureaucratically, they align disparate actors around shared visions of technological leadership. Internationally, they position states as credible and capable players in a rapidly transforming strategic landscape. The signal is therefore simultaneously inward- and outward-facing.

Strategic narratives enable states to frame their technological advancements as aligned with broader national values and goals. For instance, a state may portray its quantum technology initiatives as essential for national security and economic prosperity, thereby justifying investments and policy decisions. In strategic competition, narratives are employed to legitimise actions and delegitimise adversaries. By controlling technology narratives, states seek to shape the strategic environment to their advantage. In the case of quantum technologies, a state could frame its advancements as peaceful and beneficial for humanity, while portraying competitors' efforts as threats, thus seeking to influence international norms and policies in its favour. As such, narratives are not incidental to strategy but are co-constitutive of it. They allow states to transform speculative technologies into national projects, and they frame the metrics by which success or failure will later be judged.

By bringing anticipatory governance and strategic narrative into conversation, this paper offers a hybrid lens for analysing national technology strategies. Anticipatory governance highlights how states seek to manage uncertainty and exert control over

¹In this paper, I use “narrative” to refer to structured representations of agency and purpose deployed by states to render uncertain futures legible and governable. Following the strategic narrative literature, I distinguish these from ad hoc discourses or rhetoric; narratives are cohesive storylines embedded in official strategies that define roles (leader, follower), plot trajectories (from lagging to leading), and allocate meaning to technological change. They are not fictions but world-making devices.

potential futures. Strategic narrative, by contrast, foregrounds how states seek to shape meaning, roles, and legitimacy through techno-storytelling. Together, these perspectives illuminate how quantum strategies operate both as instruments of strategic control and as symbolic artefacts through which states narrate their place in a transforming world order.

3.4 Component 3: Status-Seeking Behaviour

Integrating status-seeking into the triadic model allows for discerning and interpreting the underlying motivations in state actions, particularly in the context of emerging technologies. By analysing national quantum strategies, the ways in which states manoeuvre to enhance their prestige and assert their positions within the international hierarchy becomes observable. Investments in quantum computing, for example, are not solely about gaining technological superiority but also about signalling a state's commitment to progress and its capability to contribute to global advancements. Such endeavours serve as markers of state actors' savviness, agility, and forward-thinking attitude. Through these actions, they aim to reshape perceptions and gain recognition from their peers, thereby altering their position within the global status hierarchy.

By incorporating status-seeking into the model, I acknowledge that states are motivated not only by security concerns but also by the desire for recognition and prestige. This perspective provides a more comprehensive understanding of state behaviour, especially in areas where material gains are intertwined with symbolic capital. In national quantum strategies, status-seeking is visible in references to global rankings, benchmarking against the US and China, and efforts to host international centres of excellence or standardization bodies. Status-seeking intensifies the political stakes of uncertainty: quantum, in this light, becomes less a technology than a stage for prestige performance.

While the focus here is on the symbolic and performative dimensions of quantum strategies, this does not deny the significant material stakes involved. These strategies are certainly also vehicles for industrial coordination, R&D investment, and the building of industrial capacity. However, their aspirational character suggests that states do not merely respond to technological competition, they imagine and construct the arenas in which that competition takes place. Thus, the symbolic and material are co-productive rather than mutually exclusive.

3.5 Interactions Among Components

While each component is analytically distinct, they are mutually reinforcing in practice. Anticipatory governance generates the need for legitimating narratives: why should public resources be allocated to speculative technologies unless they are framed as necessary and meaningful? Strategic narratives, in turn, heighten the symbolic stakes of emerging technologies, transforming them into arenas of status competition. Status-seeking behaviour, finally, adds urgency and direction to anticipatory governance—encouraging states to move first, invest visibly, and differentiate themselves rhetorically.

This triadic logic produces a feedback loop in which uncertainty becomes productive: not a constraint on action, *but a condition that enables symbolic projection and strategic identity work*. The result is aspirational security behaviour—a distinct pattern of strategic engagement with emerging technologies, characterised by material future-claiming, meaning-making, and prestige performance.

3.6 Scope Conditions and Applicability

The triadic model is designed to explain state behaviour in technological domains characterised by high uncertainty, long time horizons, and strategic ambiguity. It applies most clearly when:

1. Technological maturity is low: The technology has not yet been widely deployed or proven operationally.
2. Strategic salience is high: The technology is perceived to have future national security implications.
3. Narrative plasticity is present: The technology's meaning and significance are still open to interpretation and contestation.
4. State capacity exists: The state has sufficient institutional, bureaucratic, and discursive infrastructure to act symbolically and strategically.
5. Multipolar competition is underway: The international system features multiple actors seeking leadership in the emerging technology under consideration.

While quantum technologies are a paradigmatic case, the framework may also apply to AGI or nanotechnology, where future weaponisation is plausible; or space-based systems, such as orbital surveillance, weapon platforms, or quantum-enhanced satellites. By contrast, the model is less applicable in contexts where technologies are already mature, strategically mundane, or dominated by clear regulatory regimes.

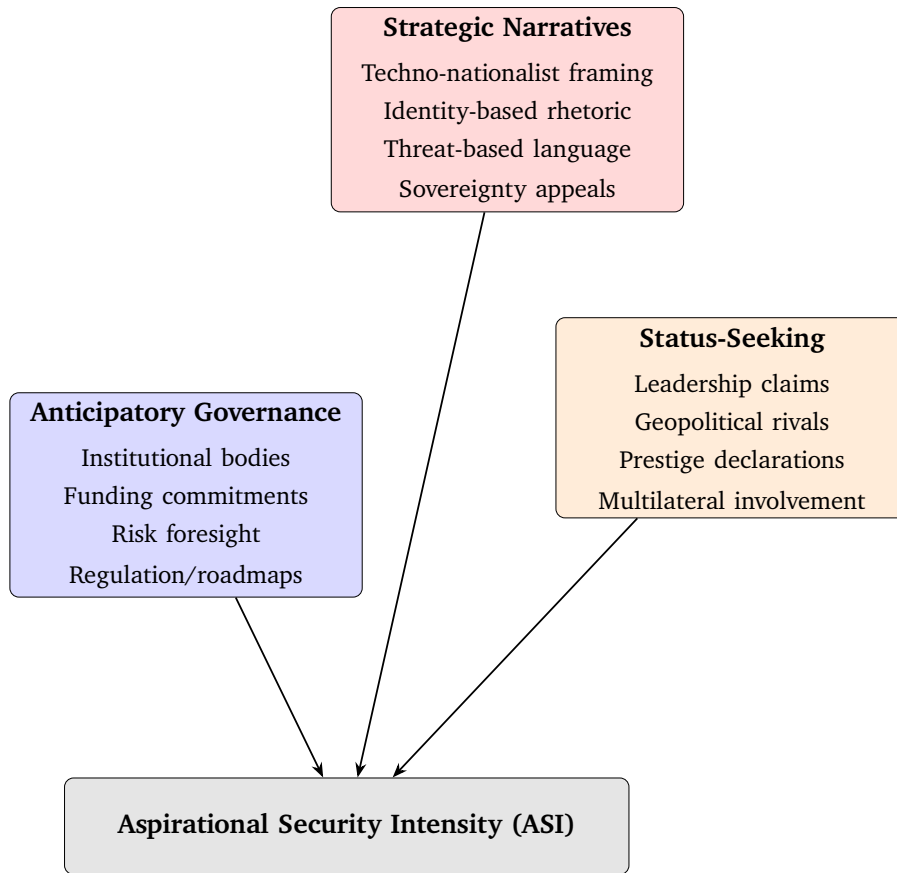
3.7 Observable Indicators and Empirical Use

In the empirical section that follows, each component is operationalised through a set of observable indicators derived from national quantum strategies. These include:

- For anticipatory governance: dedicated funding streams, long-range timelines, institutional frameworks, participation in foresight programmes,
- For strategic narratives: use of sovereignty language, leadership metaphors, identity claims, future conflict references,
- For status-seeking: benchmarking language, references to global leadership, symbolic milestones, international standard-setting roles.

These indicators guide the comparative analysis of seven national quantum strategies, revealing how states differentially combine the three components, and how variation within the triadic model helps explain different expressions of aspirational security.

The figure below illustrates the triadic framework, highlighting the three core components, their corresponding sub-indicators, and their aggregation into a composite intensity profile.



4 Research Design and Methodology

To operationalise the triadic model of aspirational security behaviour, this section introduces a comparative index-based framework for measuring the degree to which states exhibit anticipatory, narrative, and status-oriented strategic behaviour. The objective is to move beyond descriptive case analysis and provide a structured, testable means of comparing national strategies across a shared theoretical model.

4.1 Model Specification: Measuring Aspirational Security

The dependent variable is the *Aspirational Security Intensity* (ASI) of a given state's national quantum strategy. ASI is conceptualised as a function of three latent dimensions derived from the theoretical framework developed in Section 3:

- **Anticipatory Governance (AG)**: the extent to which the strategy reflects forward-looking institutional planning, long-term investment, and future-oriented policy design;
- **Strategic Narratives (SN)**: the degree to which the strategy constructs a coherent techno-political narrative around sovereignty, identity, autonomy, or conflict;
- **Status-Seeking (SS)**: the prominence of symbolic positioning, benchmarking, leadership claims, and global prestige signalling.

This relationship can be represented by a simple additive model:

$$ASI_i = \alpha \cdot AG_i + \beta \cdot SN_i + \gamma \cdot SS_i + \varepsilon_i \quad (1)$$

where:

- ASI_i is the aspirational security score for country i ;
- AG_i , SN_i , and SS_i are dimension-specific scores, derived from qualitative coding of the country's national quantum strategy;
- α , β , and γ are conceptual weights (assumed equal in the baseline model: $\alpha = \beta = \gamma = \frac{1}{3}$);
- ε_i captures unmeasured variation and noise.

In its basic form, the model does not claim to be predictive in a statistical sense, nor does it rely on large- n variance. Instead, it serves as a *structured heuristic* that allows for rigorous, replicable comparison across a focused set of states with bounded global influence. Each dimension is coded using theory-derived indicators (detailed below), and scores are combined to produce an overall ASI ranking. The results are then used to generate both qualitative insights and visualisations of comparative state behaviour under the logic of aspirational security.

This formalisation has three advantages. First, it clarifies how abstract theoretical claims can be mapped onto concrete empirical data. Second, it enables the identification of *profile variation* across cases: which states lean more heavily on narrative vs. institutional foresight vs. symbolic signalling. Third, it sets the stage for hypothesis generation and comparative typology development, offering the possibility

of distinguishing between *technocratic*, *performative*, and *hybrid* models of aspirational security behaviour.

The remainder of this section outlines the case selection, source materials, and the coding scheme used to construct the ASI index. The subsequent empirical sections apply this model to seven national quantum strategies: those of the UK, France, Germany, Canada, Japan, India, and South Korea.

4.2 Coding Scheme and Scoring Matrix

To apply the model of aspirational security behaviour to empirical cases, I construct a structured coding scheme based on the three components of the triadic framework: anticipatory governance (AG), strategic narratives (SN), and status-seeking (SS). Each component is disaggregated into four sub-indicators, derived deductively from the theoretical model and informed by preliminary readings of national quantum strategy documents.

The coding procedure assigns each sub-indicator a score on a 3-point scale:

- **0** = the feature is absent or minimally present,
- **1** = the feature is present but limited in emphasis or specificity,
- **2** = the feature is strongly emphasised, clearly articulated, and integrated into the strategic logic of the document.

This scoring approach allows for standardised comparison across national strategies and enables construction of an *Aspirational Security Index* (ASI) for each case. Each of the three dimensions has a maximum raw score of 8 (4 sub-indicators $\times$ 2). The ASI is computed as a (weighted) average of the component scores, using equal weights in the baseline model:

$$ASI_i = \frac{1}{3}(AG_i + SN_i + SS_i) \quad (2)$$

This structured coding procedure enables two levels of analysis: first, *within-case profiling* of each state's emphasis on different components of aspirational security; second, *cross-case comparison* of overall intensity and strategic orientation.

The below presents summary tables for each of the three dimensions and the empirical indicators they invite:

Table 1. Sub-indicators for Anticipatory Governance (AG)

Code	Sub-Indicator		Low (0)	Moderate (1)	High (2)
AG1	Strategic horizon	planning	≤ 2 years or not stated	3–5 years	≥ 10 -year vision
AG2	Dedicated institutions / coordination	quantum / central	None	Indirect / multi-agency	Dedicated national office or task force
AG3	Quantum funding and investment commitment		Minimal or bundled	Moderate, time-bound	Large-scale, long-term ($\geq \text{€}1\text{bn}$ or ≥ 10 years)
AG4	Future-use scenarios (e.g., cryptography, sensing)		Vague / generic	Referenced but not detailed	Clear future application roadmap

Table 1: Scoring Criteria for Anticipatory Governance (AG).

Table 2. Sub-indicators for Strategic Narratives (SN)

Code	Sub-Indicator		Low (0)	Moderate (1)	High (2)
SN1	Sovereignty / language	autonomy	Absent	Brief mention	Central narrative theme
SN2	National framing	identity	Absent	Implicit values	Explicit identity-linked claims
SN3	Reference to threats / conflict	future	Absent	General references	Linked to deterrence or future warfare
SN4	Value-based justification (e.g., democracy, ethics)		Absent	Generalised	Explicit normative mission

Table 2: Scoring Criteria for Strategic Narratives (SN).

Table 3. Sub-indicators for Status-Seeking (SS)

Code	Sub-Indicator	Low (0)	Moderate (1)	High (2)
SS1	Benchmarking vs. great powers (e.g., US/China)	None	Mentioned occasionally	Central comparator / motivator
SS2	Claims of leadership (e.g., “superpower,” “top 3”)	Absent	Vague ambition	Strong, repeated, quantified goals
SS3	Global positioning (standards, alliances)	No reference	General cooperation	Hosting or leading international efforts
SS4	Prestige metrics (rankings, KPIs, targets)	None	Some numeric goals	Multiple quantifiable global comparisons

Table 3: Scoring Criteria for Status-Seeking (SS).

4.3 Case Selection and Comparative Strategy

Cases are selected on the basis of the seven countries’ active engagement with quantum development, availability of official strategy documents, and their position outside the bipolar technological rivalry between the US and China. While these countries certainly vary in regional and institutional characteristics, all seven states face structural constraints in terms of relative material power. They therefore provide fertile ground for observing variation in aspirational security behaviour where discursive projection and institutional signalling are sought to offset the relative lack of hard-power capabilities.

The cases are analysed using a structured, theory-driven comparative design. The strategy documents are treated as the primary unit of analysis, reflecting each state’s codified expression of its goals, institutions, and identity in the quantum domain. Each document is scored using the framework outlined above, allowing for both within-case profiling and cross-case comparison. This approach facilitates identification of both common strategic logics and national variation in the emphasis placed on anticipatory governance, strategic narratives, and status-seeking.

4.4 Data Sources and Coding Procedure

Each document was accessed through official government portals or national innovation agencies. In cases where multiple relevant documents were published—such as updates, roadmaps, or related agency reports—these were reviewed and cross-referenced, but the coding was anchored in the most authoritative and comprehensive strategy document(s) per country.

All documents were either originally published in English or translated into English using official government translations. For France, Germany, Japan, and South Korea, where the primary document was issued in the national language, English-language versions released by the relevant ministry or innovation agency were used. In cases where no official translation was available (e.g., Japan's METI report), machine translations were used.

The coding process employed a structured qualitative content analysis based on the scoring framework presented above. Each document was reviewed line-by-line and scored manually on twelve sub-indicators (four each for anticipatory governance, strategic narratives, and status-seeking). Coding was conducted using a combination of spreadsheet-based scoring matrices and memo-structured qualitative annotations. All documents were read in full by the author, ensuring internal consistency across cases.

To cross-check reliability, software-assisted natural language processing (NLP) was used in order to derive summary values of term frequency and sentiment for external validation. The manual coding approach was selected to ensure contextual nuance, particularly given the diversity of rhetorical styles and political cultures across cases. Coding memos were used to record borderline judgments and scoring rationales, with particular attention paid to expressions of sovereignty, framing of technological futures, and references to global positioning ([Bryman 2012](#), [Creswell & Creswell 2018](#), [Saldaña 2015](#)).

The resulting score profiles were used to compute values for each of the three dimensions and to derive an overall *Aspirational Security Index* (ASI) for each case. These scores serve as the basis for the comparative visualisations and cross-case interpretation in Section 5.

4.5 From Framework to Model: A Theory of Aspirational Strategy Formation

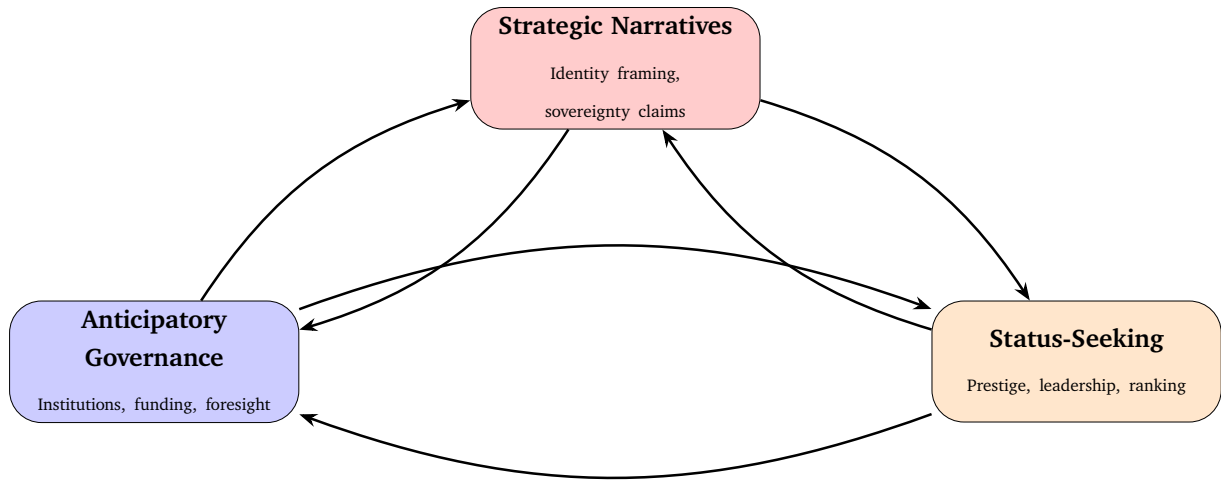
While the triadic framework developed above offers a structured method for identifying and comparing aspirational security behaviour, its core analytical potential lies in its transformation into a *dynamic model*—one that explains not only how states express anticipatory governance, strategic narratives, and status-seeking in the present, but how these elements *evolve, interact, and amplify* each other over time.

The central theoretical claim of this model is that aspirational security is not a static posture, but a *strategic process*: a sequence of moves and countermoves through which states construct, project, and defend their position in an emerging technological order. These moves are not random but follow intelligible patterns, shaped by domestic capacity, perceived strategic opportunity, and external signalling environments.

4.5.1 Interaction Effects and Strategic Feedback

In the domain of not-yet technologies, where material capabilities are absent, applications speculative, and temporal horizons long, states operate within a malleable symbolic space. I thus theorise the components of aspirational security as dynamically interdependent. All three components influence one another in context-specific ways: a narrative of quantum sovereignty may drive institutionalisation and international positioning; conversely, elite pursuit of global prestige may retroactively shape both funding structures and national framings.

Rather than fix an implausible universal causal order, the model allows for multiple “entry points” and interaction sequences. It functions as an open system of mutually recalibrating forces that co-produce aspirational security claims under conditions of technological immaturity. The diagram below visualises this dynamic interplay, while the empirical section identifies dominant logics in specific state strategies.



4.5.2 Ideal Types and Strategic Pathways

Based on observed combinations of the three dimensions, I propose a typology of *ideal-typical aspirational strategies*, each with distinct logics and vulnerabilities:

Type	AG	SN	SS	Logic	Risk
Technocratic Strategist	✓	✗	✗	Build capacity before projecting identity	Politically invisible
Sovereignty Narrator	✗	✓	✗	Lead with identity to mobilise legitimacy	Thin institutional base
Prestige Competitor	✗	✗	✓	Lead with recognition to shape standards	Symbolic overreach
Balanced Aspirant	✓	✓	✓	Synthesise capacity, meaning, and position	Resource-intensive but resilient

Table 4: Typology of Aspirational Strategy Profiles.

In summary, the above table suggests:

- **Technocratic Strategist:** High on anticipatory governance, low on narrative and status dimensions. These states prioritise capacity-building and institutional readiness without engaging in symbolic or geopolitical projection.

- **Sovereignty Narrator:** Emphasises strategic narratives while underperforming on governance and prestige. Here, identity and legitimacy claims outweigh technocratic coordination or international benchmarking.
- **Prestige Competitor:** Performs highly on status-seeking, but lacks robust governance structures or symbolic narratives. These states seek to position themselves geopolitically without necessarily backing claims through domestic institutional investment or identity work.
- **Balanced Aspirant:** Scores highly across all three dimensions. These actors build institutional capacity, construct strategic meaning, and pursue international recognition in tandem.

This typology does not seek to convey that quantum ambitions are static. States may transition from one mode to another as external conditions change (e.g., a tech breakthrough by a rival), or as domestic capacities evolve.

4.5.3 Hypotheses for Future Testing

Finally, the model generates a set of hypotheses that can guide empirical testing in future research once sufficient data become available that can speak to the actual material impact of national quantum strategies, for instance:

- **H1:** States with high strategic narrative scores are more likely to adopt anticipatory governance mechanisms within five years.
- **H2:** States that rank themselves against global powers are more likely to increase public investment in quantum technologies.
- **H3:** Balanced strategies (high AG, SN, and SS) correlate with sustained leadership in global quantum collaborations.

While this article does not test these hypotheses quantitatively, it uses qualitative scoring across seven national strategies to evaluate their plausibility and lay the groundwork for future comparative or longitudinal analysis of this kind.

5 Results and Discussion: Patterns of Aspirational Security Performance

This section presents the results of a comparative analysis of seven national quantum technology strategies using the triadic model of aspirational security. Rather than offering a case-by-case narrative, I structure the discussion around the three core dimensions of the framework: anticipatory governance, strategic narratives, and status-seeking behaviour. This approach enables a direct assessment of how states perform aspirational security through different combinations of institutional investment, symbolic projection, and international positioning. The Appendix offers a detailed, metrics-based country-by-country analysis.

I begin by offering a high-level overview of aggregate scoring trends across the triadic dimensions. This is followed by a dimension-by-dimension analysis that explores variation in how states engage each component of the model. I then map the typology of strategy profiles based on countries' composite performance across the framework. Finally, I revisit the dynamic model proposed in Section 4, evaluating the causal interplay of dimensions and reflecting on the broader implications for international security in the age of not-yet technologies.

5.1 Overview of Aggregate Trends

The aggregate scores across the three dimensions—anticipatory governance (AG), strategic narratives (SN), and status-seeking (SS)—reveal several notable patterns (see Table 5). First, all seven states perform strongly on anticipatory governance, with an average score of 7.0 out of a possible 8, reflecting widespread investment in forward-looking institutional structures and funding mechanisms. In contrast, performance on strategic narratives is more varied, with an average score of 5.1 and several cases exhibiting minimal use of symbolic or identity-based rhetoric. Similarly, status-seeking behaviour averages 5.0, with certain countries (e.g., the UK and Japan) exhibiting high ambition and international benchmarking, while others (e.g., Germany, South Korea) show more restrained signalling.

These findings suggest that while the governance of quantum technologies is broadly institutionalised across globally influential states, the symbolic and geopolitical dimensions of aspirational security remain more differentiated. This suggests that

anticipatory governance constitutes the “low-hanging fruit” of aspirational behaviour—technocratically feasible and politically less contentious—whereas symbolic projection and prestige competition require stronger narrative cohesion and geopolitical intent.

Country	AG (0–8)	SN (0–8)	SS (0–8)
Canada	7	5	3
France	8	8	7
Germany	7	4	3
India	5	7	4
Japan	8	4	7
South Korea	6	1	3
United Kingdom	8	7	8
Average	7.0	5.1	5.0

Table 5: Aggregate Scores by Dimension Across Cases.

5.2 Dimension-by-Dimension Analysis

This section disaggregates the results across the three core components of the triadic model. Rather than focusing on national profiles in isolation, I examine how states emphasise (or de-emphasise) each dimension, allowing for comparative insights into how aspirational security is assembled from different building blocks.

5.2.1 Anticipatory Governance

Anticipatory governance emerges as the most consistently developed dimension across all seven national quantum strategies. States appear most aligned in their efforts to build institutional foresight, long-term investment mechanisms, and structured planning processes for quantum technologies.

As shown in Figure 1, three countries—France, Japan, and the UK—achieve the highest scores, each demonstrating comprehensive planning horizons (to 2030 and beyond), dedicated institutional infrastructures, significant funding commitments, and clear articulation of application roadmaps. France anchors its efforts in centralised scientific mobilisation and strategic investment under a national technology plan; Japan in industrial transformation and large-scale socio-technical systems change, and the UK in mission-oriented innovation with explicit national security hooks.

Canada and Germany fall slightly behind in terms of coordination structures. Both countries exhibit strong funding and clear roadmaps. In Germany, institutional responsibility remains somewhat fragmented across ministries and agencies. India shows the lowest anticipatory governance score among the seven, with a shorter planning horizon and less clarity around implementation mechanisms; its strategy being the shortest and least detailed of the documents analysed.

The uniform strength of anticipatory governance suggests that this dimension constitutes the “baseline” of aspirational security—an actionable and politically tractable domain for states seeking to lock in quantum technology development. Unlike strategic narratives or status-seeking, anticipatory governance can be developed relatively independently of geopolitical positioning or symbolic projection, arguably making it the preferred entry point for states navigating uncertainty without provoking geopolitical tension.

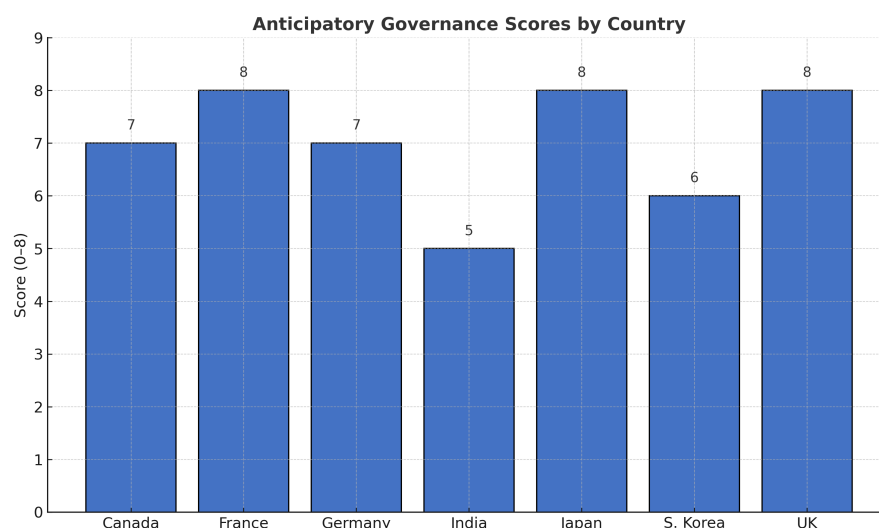


Figure 1: Anticipatory Governance Scores by Country.

5.2.2 Strategic Narratives

Compared to anticipatory governance, the dimension of strategic narratives reveals greater variation across the sample. This suggests that symbolic projection and identity work in the realm of not-yet technologies are relatively more politically and culturally contingent. Strategic narratives are harder to operationalise and standardise than governance structures, yet they serve as key signals of a state's aspirational orientation and imagined geopolitical role.

France and the UK score highest on this dimension, with strong references to sovereign capability, explicit articulation of security threats, and alignment with domestic values and national mission identity. For instance, France's strategy explicitly invokes national sovereignty and historical leadership in science and technology. India follows with moderate-to-strong narrative performance, invoking national pride and aspiration through its quantum mission. In both cases, strategic narratives are used to justify state-led mobilisation and public investment, as well as to assert technological prowess.

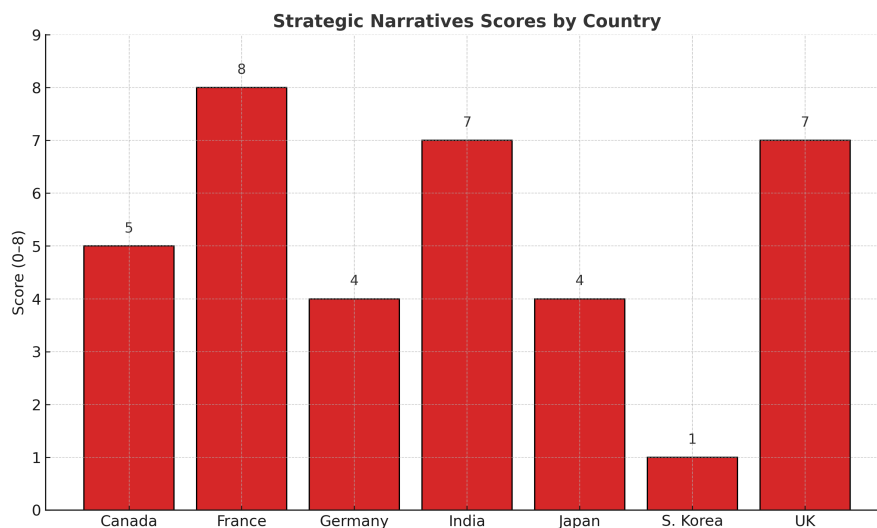


Figure 2: Strategic Narrative Scores by Country.

Canada, Germany and Japan perform moderately, incorporating value-based justifications and some identity signalling (e.g., societal wellbeing, ethical safeguards), but avoiding overt security language or sovereignty claims. Germany's strategy emphasises industrial coordination and research policy, but lacks symbolic framing beyond EU alignment. South Korea, by contrast, exhibit minimalist narrative profiles. Its strategy is more technocratic relative to other documents, prioritising

capacity-building without appealing excessively to national identity, values, or adversarial threat framing.

This pattern suggests that strategic narratives are more selectively deployed and culturally bounded or specific. While some states frame quantum technologies in grand, even existential terms, others approach them level-headed as neutral tools of economic development and societal progress. This variation is difficult to explain in terms of material capacity alone. Rather, it reflects differences in political culture, security discourse, and the perceived necessity of narrative legitimisation in the context of not-yet technologies.

5.2.3 Status-Seeking Behaviour

The third dimension—status-seeking behaviour—sits analytically between the technocratic pragmatism of anticipatory governance and the more culturally embedded realm of strategic narratives. It captures how states symbolically position themselves within global hierarchies of technological power, typically by benchmarking against great powers, articulating leadership ambitions, or signalling international relevance through prestige metrics and multilateral engagement.

The UK emerges as the highest scorer, explicitly naming the US and China as benchmarks, claiming top-three leadership goals in quantum computing, and anchoring itself in global alliances and export strategies. France and Japan also score highly, Japan presenting a comprehensive internationalisation plan and identifying clear strategic rivals.

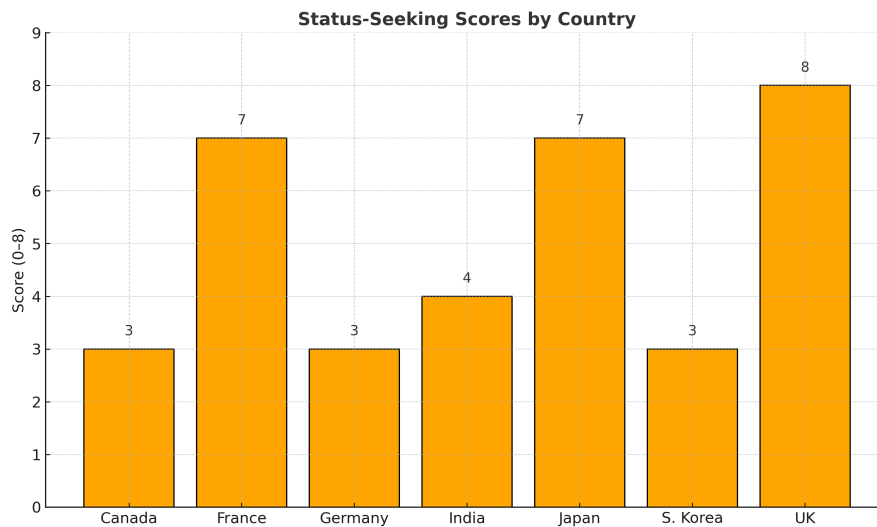


Figure 3: Status-Seeking Scores by Country.

Canada, Germany, and South Korea score more modestly on this dimension. Canada projects itself as a trusted partner in multilateral R&D, but avoids strong language around leadership or rivalry. Germany acknowledges EU-level competitiveness but refrains from national prestige framing. South Korea couches its ambition within colourful language of wanting to “stand tall as the nexus of the global quantum economy by 2035” but avoids explicit geopolitical positioning.

This variation suggests that status-seeking is a key differentiator among aspirational strategies. It is not a function of capability, but a performative act: a way of enacting relevance and projecting intent. Where anticipatory governance builds platforms, and strategic narratives construct meaning, status-seeking makes visible a state’s desired place in global hierarchies of technological prestige.

5.3 Cluster Typology of Strategy Profiles

Figure 4 visualises the dimension-by-dimension variation in state performance across the triadic model using a heatmap format. The yellow-to-red gradient highlights the relative intensity of aspirational security behaviours in each category, enabling a clear comparative scan across countries and dimensions. While most states show strong anticipatory governance foundations, strategic narratives and status-seeking behaviours vary significantly.

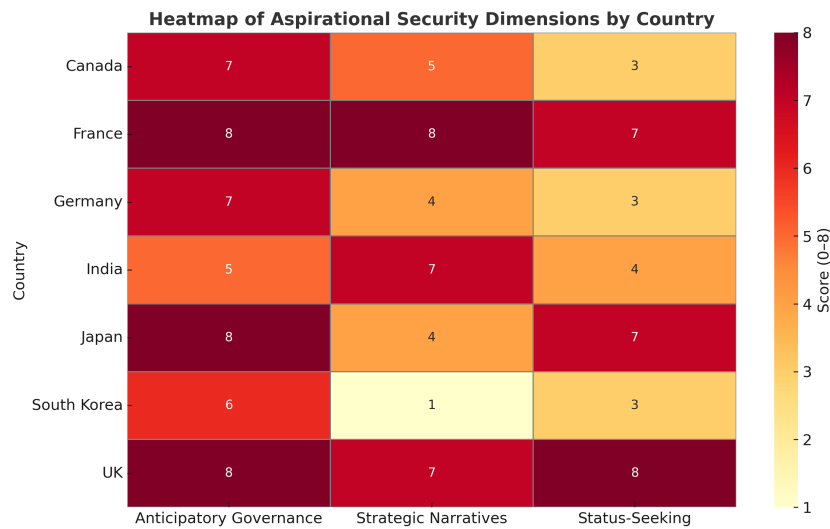


Figure 4: Heatmap of Aspirational Security Dimensions.

Figure 5 presents the same data in radar format, mapping the distribution of emphasis for each country across the three dimensions. This visualisation reveals national profile “shapes”—from the broad, balanced configurations of the UK and France to the narrower profiles of Germany and South Korea. The radar plot helps to intuitively grasp the internal balance (or imbalance) of strategic orientation and highlights which states approach aspirational security in comprehensive versus selective terms.

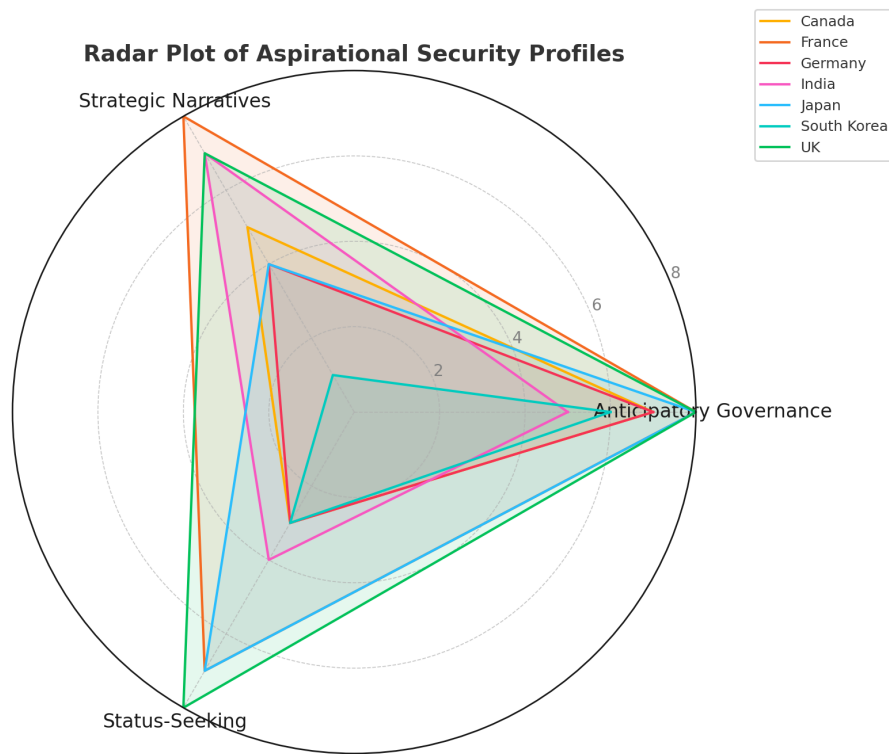


Figure 5: Radar plot of Security Profiles.

To synthesise the findings of the triadic analysis, in the previous section this article has introduced a typology of aspirational strategy profiles that captures how different states combine—or refrain from combining—the three dimensions of anticipatory governance (AG), strategic narratives (SN), and status-seeking (SS). These ideal types reflect the structural variation observed across cases and help illuminate the diversity of the strategies employed by states with bounded global influence in the quantum technology space.

Table 6 maps each country to its respective type, based on its aggregate triadic profile. While the typology is stylised, it reflects empirically observable emphases and differences in how states are positioning themselves with respect to not-yet technologies such as quantum.

Country	AG	SN	SS	Profile Type
Canada	7	5	3	Technocratic Strategist
France	8	8	7	Balanced Aspirant
Germany	7	4	3	Technocratic Strategist
India	5	7	4	Sovereignty Narrator
Japan	8	4	7	Silent Competitor
South Korea	6	1	3	Technocratic Strategist
United Kingdom	8	7	8	Balanced Aspirant

Table 6: Typology of Aspirational Strategy Profiles.

With a view to Europe and the above discussion of “technological sovereignty”, the UK’s 2023 *National Quantum Strategy*, updated in 2024, seeks to position the country as a “Science and Technology Superpower” by 2030. This ambition blends post-Brexit narratives of technological sovereignty and independence with strong anticipatory governance, including a £2.5 billion commitment over ten years and a centralised implementation structure via the Office for Quantum. While identity framing centres on autonomy and global competitiveness, particularly in declaring Europe the central competitor, the strategy is also underpinned by a bottom-up R&D model that supports startups and universities over large industrial champions. Its high scores across all three dimensions mark the UK as a *Balanced Aspirant*—a case where symbolic projection, capacity-building, and prestige-seeking are closely synchronised.

France’s 2021 strategy reflects a more centralised and sovereignty-led approach. Its €1.8 billion funding envelope is channelled through national champions like Atos and Thales, with selective support for startups. Strategic quantum narratives are explicitly geopolitical, invoking deterrence and military competitiveness. While it supports EU-wide initiatives, France does so from a position of intended leadership, folding European goals into its national vision: by aspiring to become the world’s leading quantum power, France intends to lift up Europe as a whole. This top-down orientation, combining defence imperatives with institutional depth and visible prestige claims, aligns France with the *Sovereignty Narrator* type—leading with identity and security framing while backing it with powerful governance instruments.

Germany's 2023 *Quantum Action Plan* adopts a more technocratic and decentralised approach that reflects its complex federal structure. While scoring highly on governance—due to significant funding, institutional mapping, and European coordination—the strategy avoids nationalistic narratives, instead framing its quantum ambitions as coupled “German and European” objectives. Unlike France or the UK, it lacks strong symbolic projection or status signalling, and is significantly less precise in setting measurable targets. Germany's federal structure diffuses implementation responsibility across various federal and state-level agencies, and while the plan signals strategic commitment, its four-year horizon and vague performance indicators suggest a cautious, somewhat avoidant and risk-averse posture. It exemplifies the *Technocratic Strategist* type, focused on capacity-building without a unifying aspirational identity.

While most cases align closely with one of the four ideal types of aspirational strategy, Japan presents a notable exception. With high scores in anticipatory governance and status-seeking, but only moderate emphasis on strategic narratives, Japan does not neatly fit into the existing typology. Rather than projecting a strong techno-nationalist or sovereign identity narrative, Japan's strategy foregrounds institutional coordination and global leadership claims, while remaining relatively subdued in symbolic or value-laden framing. This hybrid configuration—combining technocratic competence with prestige ambition but minimal rhetorical assertion—may reflect Japan's diplomatic style and domestic political culture, which favour global positioning through quiet expertise rather than noisy signalling (Dang & Glenn 2022, Solis 2023).

The classification above highlights a key insight of the study: while anticipatory governance provides the most widely adopted foundation, very few states pursue a fully balanced aspirational strategy. The UK and France are notable exceptions, going all in on aligning governance, narrative, and prestige. Other countries opt for selective engagement, arguably shaped by domestic constraints and geopolitical sensitivities. The typology thus serves as a heuristic for placing and interpreting national strategies not only in terms of capacity-building, but also regarding the performative and symbolic dimensions of quantum technology development.

In these assemblages, the tension between the state as uncertain and the state as capable to deliver on its quantum promises is not a contradiction but a constitutive feature of anticipatory governance. Precisely because future technological leadership cannot be guaranteed, states engage in performances that simulate certainty—projecting coordination, control, and epistemic authority even when

technological progress remains fragmented and volatile. These strategies thus operate as both managerial instruments and symbolic stabilisers in the face of ambiguity.

5.4 Interpreting Patterns of Variation

The analysis above suggests that states do not converge around a singular model of aspirational security. Instead, they mobilise different combinations of governance, narrative, and prestige to signal relevance and build legitimacy in the domain of quantum technologies, reflecting distinct national constraints, political traditions, and geopolitical orientations.

One key pattern is that anticipatory governance emerges as a broadly shared baseline. All states—regardless of their narrative or prestige posture—invest in future-oriented planning, institutional coordination, and public funding. This suggests that anticipatory governance is the most politically tractable and least contested form of aspirational behaviour. It allows states to appear forward-looking and strategically competent without overtly engaging in geopolitical signalling or value-laden discourse.

By contrast, strategic narratives and status-seeking are more selectively deployed. Strategic narratives tend to be stronger in countries with a history of national mission framing (UK, France, India), and weaker in states with consensus-driven or export-led innovation cultures (Germany, South Korea). Status-seeking, meanwhile, correlates with either strong global ambition (UK, Japan) or explicit leadership claims as a form of soft power leverage (France, India). Yet some countries—like Canada and Germany—opt for modesty or multilateral positioning rather than headline-grabbing declarations.

These differences are unlikely to be static. As the quantum field matures, states may shift emphasis across dimensions, adding narrative to governance, or seeking prestige once basic capabilities are secured. While the triadic model in its basic form offers a snapshot of current behaviours, it may also serve a template for tracking how aspirational security strategies evolve over time.

5.5 Implications for Theory

The findings of this study contribute to broader theoretical debates in international relations and security studies in three ways.

First, the analysis demonstrates that strategic behaviour in emerging technologies is not solely a function of material capabilities or immediate threat perceptions. Rather, it often unfolds as a future-oriented practice of meaning-making, legitimacy construction, and identity projection. This insight reinforces the need to complement realist and rationalist frameworks with approaches that take temporality, discourse, and symbolic agency seriously.

Second, the triadic model shows that anticipatory governance, strategic narratives, and status-seeking behaviour are not mutually exclusive, but co-constitutive components of how states perform aspirational security in the domain of not-yet technologies. The model helps theorise how states navigate the “pre-material” phase of technology competition before capabilities are fully developed, battlefield-ready, or economically dominant. This study shows that technology can function as a narrative object and status arena long before its operationalisation. This anticipatory mode of statecraft is especially visible in aspirational powers with bounded global influence that punch under the weight of great powers in terms of their material capabilities, and whose strategic presence often leans on symbolic projection rather than hard power.

Taken together, these theoretical implications suggest that emerging technology governance is not merely reactive or about “regulating-after-the-fact” ([Issacharoff 2007](#)), a common charge against practitioners in governance and regulation, exacerbated by the speed of development and progress in evolving domains such as AI ([Judge et al. 2024](#)). It is also a site of strategic imagination, where states articulate visions of technology futures to gain leverage in the present.

6 Conclusion

This article began with a puzzle: how do states compete over technologies that do not yet exist? In response, this paper has introduced a new conceptual and analytical model for understanding how states engage in strategic competition over emerging technologies that remain largely undeployable. Through the triadic lens of anticipatory

governance, strategic narratives, and status-seeking behaviour, I proposed the concept of *aspirational security*—a form of state practice in which the future itself becomes a domain of geopolitical contestation. A structured content analysis of seven national quantum strategies demonstrated that countries below the great-power threshold do not passively follow the lead of technological hegemons. Rather, they actively project agency through symbolic, institutional, and narrative means to claim stakes in desired technology futures. They do this to shape the future before it arrives.

The model developed here enables a richer account of technological statecraft, one that moves beyond traditional capability-based approaches to integrate the performative and institutional dimensions of strategic behaviour. Importantly, the findings show that while anticipatory governance enjoys a high baseline across cases, states vary considerably in how they narrate their technological identities and position themselves globally. These variations reflect underlying national logics shaped by political culture and institutional design, which I categorised through a typology of aspirational strategy profiles.

A natural question follows: why do states not always behave in this triadic manner toward all technologies? The answer lies in the particular conditions of emerging, not-yet-realised technologies. When material capabilities are still under development, uncertainty is high, and timelines are speculative, the symbolic and anticipatory dimensions of strategy become more salient. In such settings, narrative projection, institutional signalling, and status performance are not auxiliary, they are constitutive of power and expressions thereof. In contrast, mature technologies offer less space for such symbolic manoeuvre: expectations are stabilised, trajectories are path-dependent, and competition centres on deployment, not definition. The triadic model thus applies most forcefully under conditions of immaturity, where the meaning, purpose, and ownership of not-yet technologies remain contested and open to strategic interpretation.

This article has demonstrated that national quantum strategies do not merely coordinate research funding; they perform sovereignty, stage authority, and cast states as protagonists of the future. In doing so, they reveal the aspirational dimensions of contemporary security governance—where power is not just exercised, but narrated through desired technology futures. By reading these strategies as aspirational security narratives, a clearer view is gained of how states navigate uncertainty and geopolitical flux. Far from being technocratic footnotes, these documents are critical sites where futures are invented, the techno-identities of state actors rehearsed, and power claims

legitimated.

As competition over not-yet technologies intensifies, across quantum and beyond, understanding the aspirational dimensions of strategic behaviour becomes increasingly urgent. In an era when the future itself is a geopolitical arena, the capacity to anticipate, narrate, and institutionalise tomorrow's capabilities has become a significant source of power in the present.

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Appendix: Country-Level Findings and Scoring Matrices

This appendix provides detailed country-by-country findings based on a structured content analysis of seven national quantum strategies. For each case, I present a scoring matrix that captures the degree of aspirational security exhibited across the three dimensions of the triadic model: anticipatory governance (AG), strategic narratives (SN), and status-seeking (SS). Each dimension is assessed using four sub-indicators, scored on a scale from 0 (low) to 2 (high), yielding a maximum of 8 points per dimension and 24 points overall.

The matrices are accompanied by brief analytical summaries that contextualise the strategy within the broader comparative framework. This material serves as the empirical backbone of the typological analysis presented in Section 5.

Canada

Canada considers itself a “global leader” in the the field of quantum science, enjoying a comfortable lead over its competition. However, “as the rest of the world expands its own quantum programs, Canada must continue to invest and innovate if it is to stay ahead”, the strategy notes, pointing to increased economic pressure arising from international competition. While the tone is technocratic and collaborative, Canada signals quiet ambition in its overall effort to “make Canada a world leader in the continued development, deployment and use of quantum computing hardware and software”. Overall, Canada’s quantum vision is geared primarily towards economic gain and commercialisation ([Csenkey & Graver 2024](#)). The three strategy documents reveal a deeply anticipatory posture-one that leans on structured governance more than on symbolic nationalism or overt prestige-seeking.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	2
AG2	Quantum institutions / coordination	2
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	1
SN1	Sovereignty / autonomy language	1
SN2	National identity framing	1
SN3	Reference to future threats	2
SN4	Value-based justification	1
SS1	Benchmarking vs. great powers	1
SS2	Claims of leadership	1
SS3	Global positioning	1
SS4	Prestige metrics	0

Table 7: Scoring Matrix – Canada Quantum Strategy (2021–2030).

Canada fits the **Technocratic Strategist** type in the typology developed in the main sections of the paper. Its approach is characterised by:

- A robust anticipatory governance infrastructure,
- A cautious and credible security-focused narrative,
- A lack of overt prestige signalling.

France

France’s quantum strategy is among the most ambitious and state-centric in the set. Framed as a matter of national sovereignty and global leadership, the strategy combines bold funding commitments, high-profile industrial partnerships, and a strong

sense of technological urgency. The narrative is overtly geopolitical: France positions itself as a defender of European sovereignty in the face of great-power rivalry and Big Tech hegemony. At the same time, the state asserts a role for quantum technology as an enabler of industrial renewal, defence readiness, and cyber resilience. What distinguishes the French strategy is the **tight integration of symbolic, material, and governance elements**—it constructs a national vision of technological power that draws equally on science, sovereignty, and prestige. The 2021 strategy and its 2023 progress report reflect a **high-intensity aspirational posture**, combining all three components of the model.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	2
AG2	Quantum institutions / coordination	2
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	2
SN1	Sovereignty / autonomy language	2
SN2	National identity framing	2
SN3	Reference to future threats	2
SN4	Value-based justification	2
SS1	Benchmarking vs. great powers	2
SS2	Claims of leadership	2
SS3	Global positioning	2
SS4	Prestige metrics	1

Table 8: Scoring Matrix – France Quantum Strategy (2021–2023).

France fits the **Balanced Aspirant** type. It maximally activates all three legs of the model:

- Symbolic framing of sovereignty and leadership,
- Technocratic capacity-building across sectors,

- Clear international signalling of ambition and benchmarks.

This reflects France’s tradition of state-led innovation and its aspirational role in global affairs. France’s approach is not just technocratic or performative—it is **strategically choreographed**, integrating foresight, narrative control, and geopolitical projection into a coherent whole. It is among the clearest cases of **aspirational security at full intensity** in the dataset.

Germany

Germany’s complex federal structure reflects in its quantum strategy. Unlike France’s centralised model, the German strategy reveals a federated approach to quantum development, working through Germany’s network of public research institutes (Fraunhofer, Max Planck, Helmholtz) and industry alliances. Quantum technologies are presented as both geopolitical challenge and a societal investment (e.g. for managing climate risks), requiring long-term state coordination and infrastructure building. Future security challenges are acknowledged, particularly around encryption, dependency on foreign technologies, and talent outflows. While much less overtly geopolitical than France, Germany conveys quiet, somewhat modest ambition.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	1
AG2	Quantum institutions / coordination	2
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	2
SN1	Sovereignty / autonomy language	1
SN2	National identity framing	1
SN3	Reference to future threats	1
SN4	Value-based justification	1
SS1	Benchmarking vs. great powers	1
SS2	Claims of leadership	1
SS3	Global positioning	1
SS4	Prestige metrics	0

Table 9: Scoring Matrix – Germany Quantum Strategy (2022–2023).

Germany occupies a **Balanced Aspirant** position with a technocratic tilt. Its anticipatory governance scores are maximised, its sovereignty framing is strong, and its status-seeking is evident but not theatrical. The strategy is:

- Deeply institutional,
- Quietly competitive,
- Federally orchestrated.

Germany’s strategy is an instructive case of modest signalling where showcasing of federal funding schemes replaces declarative ambition.

India

While shorter and more compact than other national strategies, the National Quantum Mission nonetheless communicates a powerful political and strategic narrative. The programme outlines ambitious goals that are framed not only as technical milestones but as national achievements; the strategy is supposed to “provide a huge boost to national priorities like Digital India, Make in India, Skill India and Stand-up India, Start-up India, Self-reliant India and Sustainable Development Goals (SDG)”, pointing to a relative priority of status narratives and signalling.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	1
AG2	Quantum institutions / coordination	1
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	1
SN1	Sovereignty / autonomy language	2
SN2	National identity framing	2
SN3	Reference to future threats	1
SN4	Value-based justification	2
SS1	Benchmarking vs. great powers	1
SS2	Claims of leadership	1
SS3	Global positioning	1
SS4	Prestige metrics	1

Table 10: Scoring Matrix – India Quantum Mission (2023).

India’s National Quantum Mission can be classified as a Sovereignty Narrator with technocratic aspirations. It leads with national identity and values-based justification, underscoring autonomy and development. It is less institutionalised than the German or French cases but features:

- An ambitious if undetailed roadmap,
- Symbolically rich, sovereignty-anchored rhetoric,
- High-level political alignment with national missions.

India exemplifies how emerging powers seek symbolic and strategic stakes in technological futures without necessarily framing themselves in direct rivalry with global quantum frontrunners. This aspirational mode is narrative-led, sovereign-facing, and infused with techno-ethos and national pride.

Japan

Rather than overt sovereignty claims, Japan positions quantum as part of a transition to a *Society 5.0*: a hyper-connected, secure, and resilient socio-technical order. The three-part strategic roadmap reflects strong anticipatory governance: clear milestones, technology readiness levels, and hybrid quantum–classical testbeds. It situates quantum technology development within national industrial policy and global supply chain resilience. Japan emphasises collaboration and competitiveness over confrontation, preferring technical leadership to political posturing, seeking continuity with early-day quantum research ([Yamamoto et al. 2019](#)). Nevertheless, international benchmarking and output-based prestige metrics are still prominent, particularly in economic projections and global positioning.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	2
AG2	Quantum institutions / coordination	2
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	2
SN1	Sovereignty / autonomy language	1
SN2	National identity framing	1
SN3	Reference to future threats	1
SN4	Value-based justification	1
SS1	Benchmarking vs. great powers	2
SS2	Claims of leadership	1
SS3	Global positioning	2
SS4	Prestige metrics	2

Table 11: Scoring Matrix – Japan Quantum Strategy (2020–2023).

Japan presents as a Balanced Aspirant, exhibiting high levels of anticipatory governance and prestige-oriented signalling, but moderate symbolic nationalism. Its strength lies in:

- Institutional orchestration and roadmap detail,
- Strong international benchmarking and economic ambition,
- Moderate identity narratives and restrained security language.

Japan offers a model of aspirational security anchored less in sovereignty discourse and more in industrial policy and socio-technical innovation. The narrative is one of global contribution, prestige through metrics, and resilience through planning.

South Korea

The South Korean quantum strategy is anchored in the logic of national competitiveness and ecosystem growth, rather than symbolic projection or normative justification. The roadmap is heavy on institutional modernisation, private-sector enablement, and skills development. By and large, the narrative avoids overt sovereignty claims or strategic confrontation. Instead, the tone is technocratic and incremental, suggesting that South Korea sees quantum technologies as levers for productivity, export power, and technological independence—without over-politicising their development trajectory.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	2
AG2	Quantum institutions / coordination	1
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	1
SN1	Sovereignty / autonomy language	0
SN2	National identity framing	1
SN3	Reference to future threats	0
SN4	Value-based justification	0
SS1	Benchmarking vs. great powers	1
SS2	Claims of leadership	1
SS3	Global positioning	1
SS4	Prestige metrics	0

Table 12: Scoring Matrix – South Korea Quantum Strategy (2023).

South Korea aligns with the Technocratic Strategist profile. It prioritises:

- Capacity-building and institutional investment,
- Low political visibility or rhetorical symbolism,

- Moderate global benchmarking and future positioning.

This posture reflects a sober and infrastructural approach to quantum development grounded in competitiveness and capability, rather than symbolic projection or normative ambition.

United Kingdom

The UK presents one of the most structured, forward-looking, and security-aware national quantum strategies among the set of powers assessed. The strategy is anchored in a ten-year vision and articulates national ambitions not only in economic terms but also with a clear geopolitical undertone, preparing the grounds “for the UK to be a leading quantum-enabled economy”.

The UK strategy integrates anticipatory governance, symbolic sovereignty language, and security threat awareness into a fairly unified, strong and cohesive signal of aspirational behaviour. It is supported by infrastructure initiatives, as well as a growing set of export and defence partnerships.

Code	Sub-Indicator	Score (0–2)
AG1	Strategic planning horizon	2
AG2	Quantum institutions / coordination	2
AG3	Quantum funding commitment	2
AG4	Future-use scenarios	2
SN1	Sovereignty / autonomy language	2
SN2	National identity framing	2
SN3	Reference to future threats	2
SN4	Value-based justification	1
SS1	Benchmarking vs. great powers	2
SS2	Claims of leadership	2
SS3	Global positioning	2
SS4	Prestige metrics	2

Table 13: Scoring Matrix – United Kingdom Quantum Strategy (2023–2024).

The UK presents a strong case of a Balanced Aspirant strategy, integrating:

- Robust anticipatory governance infrastructure,
- Clear sovereignty-based strategic narrative, and
- Assertive international positioning across alliances and standards.

The strategy moves between scientific ambition and national security foresight. Compared to other powers of bounded global influence, it leans furthest into the integration of aspirational security, constructing itself as both a technological leader in quantum technologies and a normative bulwark in an increasingly adversarial world order.